

JBM9204 INTRODUCTION TO BIOMETRIC SYSTEMS

UNIT-I

BIOMETRIC SYSTEMS AND AUTHENTICATION TECHNOLOGIES

What is Biometrics?

Biometrics is the science and technology of measuring and analyzing biological or behavioral characteristics of individuals for the purpose of identification and authentication.

Biometrics=bios+metron. Bios means life and metrons means measure.

History of Biometrics

In Ancient Times Fingerprints were used in Babylon (500 BC) and China (14th century) for identification and signatures.

In 19th Century:

- Bertillon system (Anthropometry) by Alphonse Bertillon (Father of Biometrics).
- Fingerprint classification by Francis Galton.
- Henry Classification System adopted by police forces.

In 20th Century:

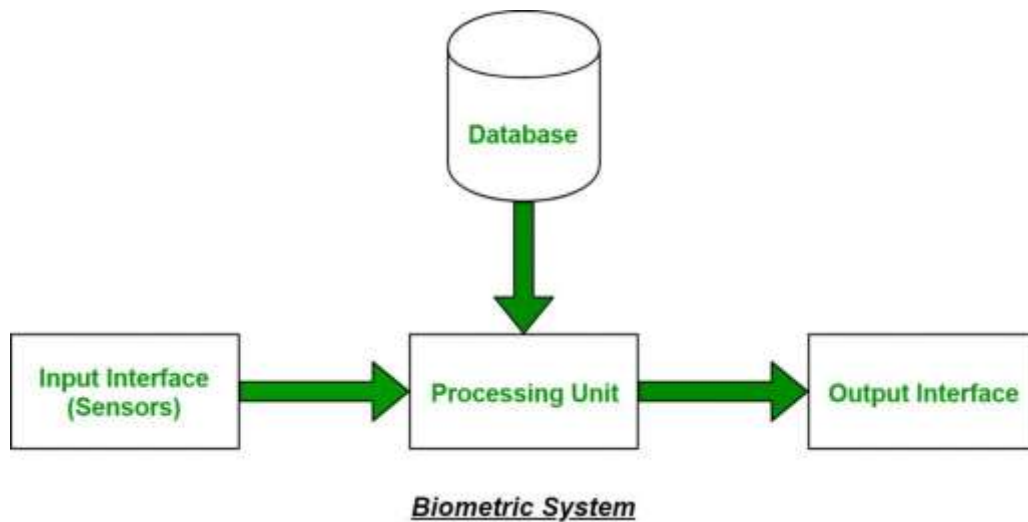
- Fingerprint identification widely used in law enforcement.
- Development of automated fingerprint and face recognition systems.
- Introduction of iris and voice recognition technologies.(John Dougman)

In 21st Century:

- Biometrics used in smartphones, border security, banking.
- India's Aadhaar project introduced large-scale multimodal biometrics.
- AI and deep learning enhanced accuracy.
- Rise of contactless biometrics post-COVID.

Biometric system

A biometric system is an automated system that captures, processes, and compares a person's unique physical or behavioral traits to recognize or confirm their identity.

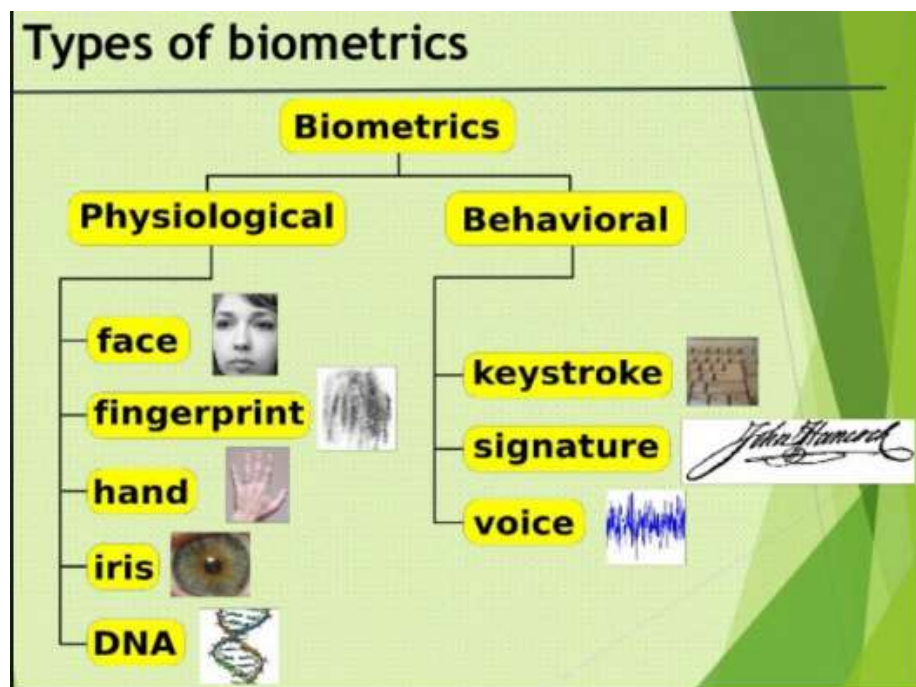


The biometric feature being used must be available in the database for all individuals in the community before the feature can be used for authentication. This is called **enrollment**.

Types of Biometrics

There are two broad categories of biometrics:

1. Physiological Biometrics (based on body traits)
2. Behavioral Biometrics (based on human actions)



Physiological Biometrics:

Physical traits are measured for identification and verification in this type of biometrics. The trait should be chosen such that it is unique among the population, and resistant to changes due to illness, aging, injury, etc.

- **Fingerprint:** Fingerprints are unique for every individual. They can be measured in several ways. Minutiae-based measurement uses graphs to match ridges whereas image-based measurement finds similarities between the individuals' fingertips image and fingerprint images present in the database. It has high level of security and used both for identification and verification. However, due to old age or diseases/injury, fingerprint may get altered.
- **Facial Recognition:** Features of the face like distance between the nose, mouth, ears, length of face, skin color, are used for verification and identification. Accuracy can be affected by fog, sunglasses, aging, etc.
- **Iris and Retina: Patterns** found in the eye are unique and can be used for both identification and recognition. Devices to analyze retina are expensive and hence it is less common. Diseases like cataract may alter iris patterns
- **DNA:** DNA is unique and persistent throughout lifetime. Thus, security is high and can be used for both identification and verification.

Behavioral Biometrics:

Traits of human behavior are measured in this case.

- **Signature:** Signature is one of the most commonly used biometrics. They are used to verify checks by matching the signature of the check against the signature present in the database. Signature tablets and special pens are used to compare the signatures. Duration required to write the signature can also be used to increase accuracy. Signatures are mostly used for verification.
- **Voice Recognition:** The pitch, voice modulation, and tone, among other things are measured. Security is medium, due to the similarity in voice of people, hence used mostly for verification. The accuracy can be hindered due to the presence of noise, or due to aging or illness.
- **Keystroke Dynamics:** This technique measures the behavior of a person when typing on a keyboard. Some of the characteristics take into account are:
 - Typing speed.
 - Frequency of errors
 - Duration of key depressions

Criteria for selection of Biometric:

- **Universality:** Each person should possess the biometric trait which is being used. For example, everyone has a face, but it is not the case with GAIT biometric (for wheelchair users).
- **Uniqueness:** No two persons must be same in terms of the biometric trait being used i.e. everyone must be unique in terms of the biometric trait being used.
- **Permanence:** Biometric trait must be invariant over time i.e. it shouldn't change over time.
- **Collectability:** Biometric trait must be easily measurable.
- **Performance:** Processing of the biometric trait must be accurate and fast.
- **Secure:** It must be secure and can't be copied.
- **Acceptability:** People should be willing to accept the biometric system.

BIOMETRIC IDENTIFICATION, VERIFICATION AND AUTHENTICATION

Biometric Identification

This process determines the individual's identity by comparing their biometric data against a database of stored biometric information. It answers the question, "Who is this person?" It involves a one-to-many (1: N) matching system, where the biometric sample is compared to multiple records in the database to find a match. It's commonly used in border control and law enforcement.

Purpose: To identify an unknown individual.

Process: One-to-many (1: N) matching.

In law enforcement, organizations utilize biometric identification, such as facial recognition or fingerprint databases when identifying suspects in criminal investigations. This technology allows for the quick identification of individuals from a large database of known criminals or suspects, ultimately assisting in resolving crimes, so long as the law enforcement body has access to a prior record of the individual being searched.

Biometric Verification

Biometric verification is a process that confirms an individual's claimed identity by comparing their biometric data to a specific stored record. It's like answering the question, "Are you who you say you are?" This process uses a one-to-one (1:1)

matching system, where the biometric sample is compared to a single stored record to verify the claimed identity. It's commonly used for access control, such as financial transactions, demonstrating its practical relevance.

Purpose: To verify a claimed identity.

Process: One-to-one (1:1) matching.

In the banking industry, it is reasonable for banks to implement biometric verification during high-value transactions or at ATMs to enhance security and prevent fraud. This extra layer of security helps confirm the individual's identity before allowing financial operations to proceed.

Biometric Authentication

Biometric authentication evaluates the genuineness of an individual as a method of confirming their authorization to access the system or specific information. It involves establishing trust in the authenticity, validating a person's identity and an authenticator.

Purpose: To confirm an individual's identity.

Process: One-to-one (1:1) matching.

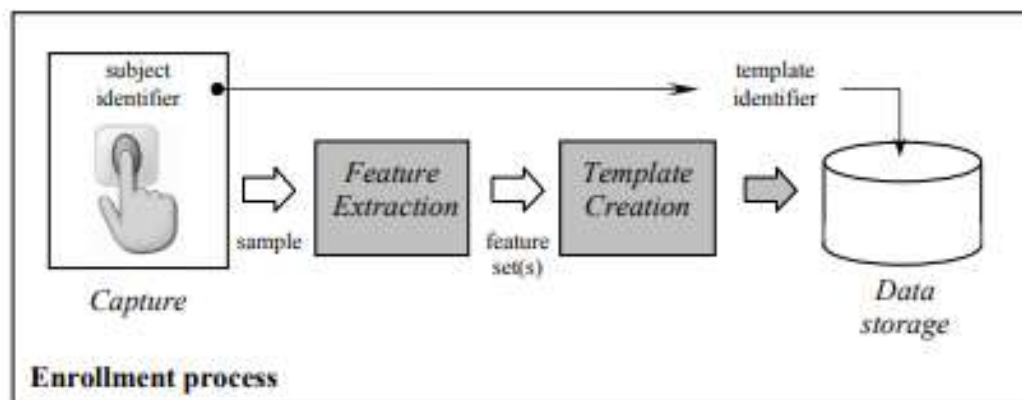
Sometimes, companies may add one-to-many (1:N) checks to their authentication workflow, like against a blacklist of known fraudsters, but authentication itself is a one-to-one (1:1) process.

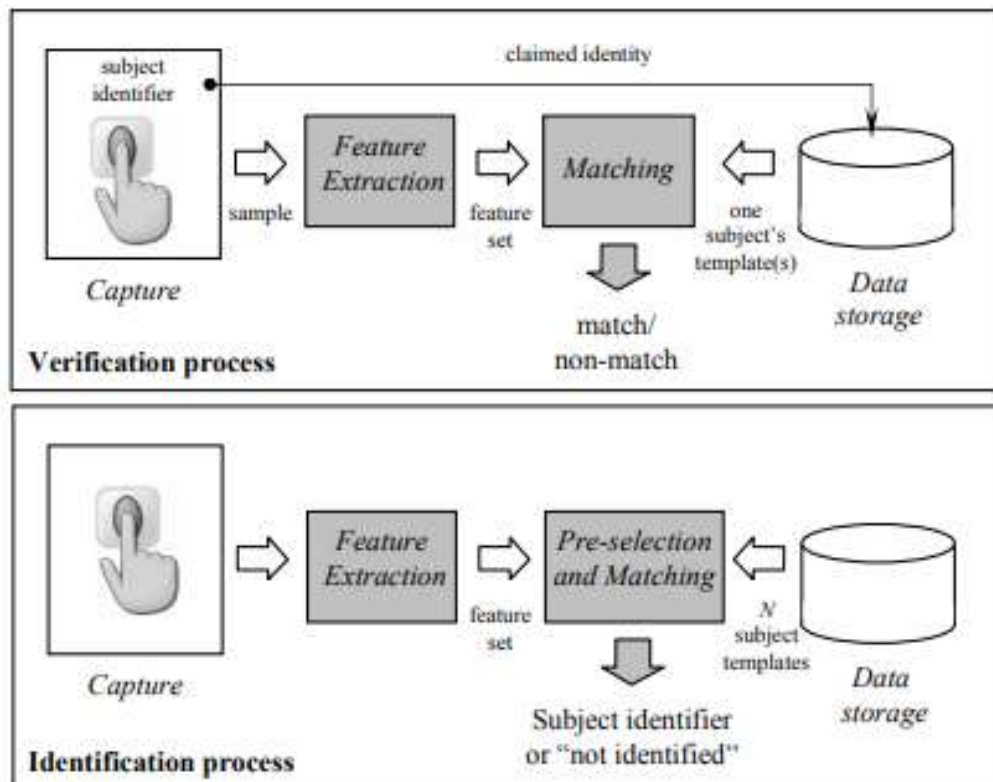
Biometric authentication is commonly observed when a user unlocks a smartphone using fingerprint or facial recognition. In this context, the application necessitates a secure method to authenticate the user's asserted identity, thus ensuring that only the authorized user can access the device.

Verification vs Authentication

Aspect	Verification	Authentication
Definition	Confirms accuracy of claimed identity	Confirms identity of a person or entity
Purpose	To check "Are you really who you say you are?"	To determine "Who are you?" or verify access
Process Type	One-to-One (1:1) matching	Broader process, can include 1:1 or 1:N matching

User Action	User claims an identity	User may or may not claim an identity
Example (Biometric)	Matching a fingerprint to one stored template	Using fingerprint + password to access your account
Purpose	To check “Are you really who you say you are?”	To determine “Who are you?” or verify access
Process Type	One-to-One (1:1) matching	Broader process, can include 1:1 or 1:N matching





GENERAL ARCHITECTURE OF BIOMETRIC SYSTEM AND ITS FUNCTIONALITY

Biometrics are computerized methods of identifying a person based on physiological and observable qualities. The use of biometric systems has impacted the way we identify and authenticate ourselves around the world. By utilizing this technology, not only has the identification of people changed, but also the time it takes to identify and verify people has been significantly reduced.

Phases of Biometric System

There are two phases of a Biometric System:

1. Enrollment phase:

In the enrollment phase, biometric information of the user or person is recorded in a database. It is a one-time process. Generally, in this phase, measurement of the appropriate information is done very precisely.

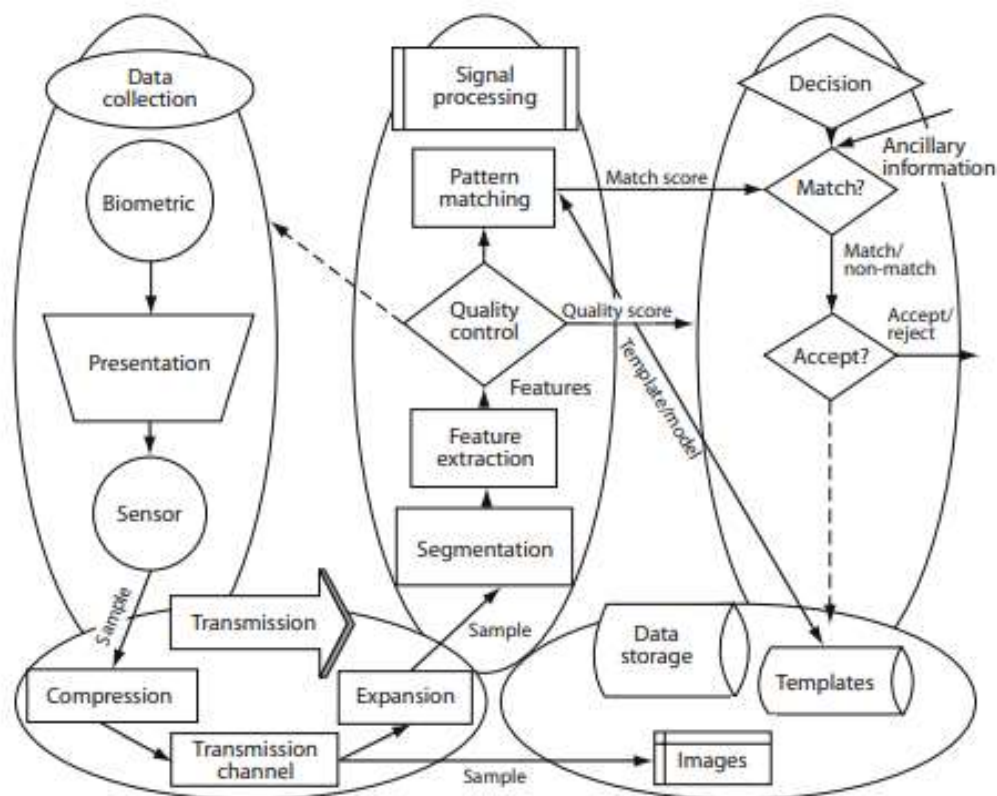
2. Recognition phase:

This is the second phase of the biometric system. This occurs when the detection part begins based on the first phase of the authentication of the user. This phase must be quick, accurate, and able to determine the authentication problem easily.

Biometric System Architecture

The below figure shows a generic biometric authentication system divided into five subsystems:

1. Data collection
2. Transmission
3. Signal processing
4. Decision and
5. Data storage



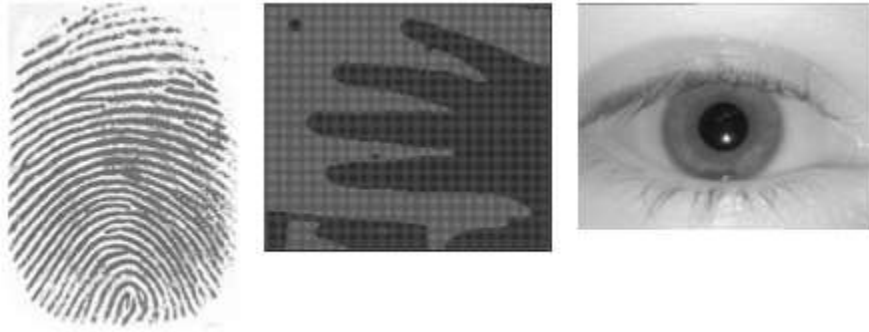
1. Data Collection

Biometric systems begin with the measurement of a behavioral/physiological characteristic.

The user's characteristic must be presented to a sensor. The presentation of any biometric characteristic to the sensor introduces a behavioral (and, consequently, psychological) component to every biometric method. This behavioral component may vary widely between users, between applications, and between the test laboratory and the operational environment.

The output of the sensor, which is the input data upon which the system is built, is the convolution of:

- (1) The biometric measure;
- (2) The way the measure is presented; and
- (3) The technical characteristics of the sensor.



Input to the system

2. Transmission

Some, but not all, biometric systems collect data at one location but store and/or process it at another. Such systems require data transmission. If a great amount of data is involved, compression may be required before transmission or storage to conserve bandwidth and storage space. The process of compression and expansion generally causes quality loss in the restored signal, with loss increasing with increasing compression ratio. The compression technique used will depend upon the biometric signal. Standards currently exist for the compression of fingerprints (Wavelet Scalar Quantization), facial images (JPEG), and voice data (Code Excited Linear Prediction).

3. Signal Processing

Having acquired and possibly transmitted a biometric characteristic, we must prepare it for matching with other like measures. The above figure divides the signal-processing subsystem into four tasks: segmentation, feature extraction, quality control, and pattern matching.

Segmentation

Segmentation is the process of finding the biometric pattern within the transmitted signal. For example, a facial recognition system must first find the boundaries of the face or faces in the transmitted image. A speaker verification system must find the

speech activity within a signal that may contain periods of non-speech sounds. Once the raw biometric pattern of interest has been found and extracted from larger signal, the pattern is sent to the feature extraction process.

Feature Extraction

Feature extraction is fascinating. The raw biometric pattern, even after segmentation from the larger signal, contains non-repeatable distortions caused by the presentation, sensor and transmission processes of the system. These non-controllable distortions and any non-distinctive or redundant elements must be removed from the biometric pattern, while at the same time preserving those qualities that are both distinctive and repeatable. In general, feature extraction is a form of non-reversible compression, meaning that the original biometric image cannot be reconstructed from the extracted features. In some systems, transmission occurs after feature extraction to reduce the requirement for bandwidth.

Quality Control

After feature extraction, or maybe even before, we will want to check to see if the signal received from the data collection subsystem is of good quality. If the features “don’t make sense” or are insufficient in some way, we can conclude quickly that the received signal was defective and request a new sample from the data collection subsystem while the user is still at the sensor. If a negative decision by the quality control module cannot be overridden, a “failure to enroll” error results. The development of this “quality control” process has greatly improved the performance of biometric systems in the last few short years.

Pattern Matching

The feature “sample”, now of very small size compared to the original signal, will be sent to the pattern matching process for comparison with one or more previously identified and stored feature templates or models. The purpose of the pattern matching process is to compare a presented feature sample to the stored data, and to send to the decision subsystem a quantitative measure of the comparison.

4. Storage

The remaining subsystem to be considered is that of storage. There will be one or more forms of storage used, depending upon the biometric system. Templates or models from enrolled users will be stored in a database for comparison by the pattern matcher to incoming feature samples. For systems only performing “one-to-one” matching, the database may be distributed on smart cards, optically read cards or magnetic stripe cards carried by each enrolled user. The database will be centralized

if the system performs one-to-N matching with N greater than one, as in the case of identification or “PIN less verification” systems. As N gets very large, system speed requirements dictate that the database be partitioned into smaller subsets such that any feature sample need only be matched to the templates or models stored in one partition, or indexed by using an appropriate data structure which allows the templates to be visited in an advantageous order during the retrieval.

5. Decision

The decision subsystem implements system policy by directing the database search, determines “matches” or “non-matches” based on the distance or similarity measures received from the pattern matcher, and ultimately makes an “accept/reject” decision based on the system policy.

Benefits of using Biometric system over traditional authentication systems:

- **Invariant:** Biometric traits are invariant over time as smart cards get damaged over time, but biometric traits doesn't.
- **Accountability:** If there is a security breach, then biometric ensures who can be the responsible person for the breach but in traditional methods, smart cards can be stolen and used by someone else. Hence, accountable person is easily identifiable nowadays by using biometric.
- **Easy to use:** Biometric systems are easy to use.
- **Convenient:** User doesn't have to remember passwords, pins and keep safe the smart cards like before.
- **More secure:** Biometric trait can't be stolen or copied.

Advantages of Biometrics

Listed below are some of the major advantages of using biometrics and biometric security systems –

- Biometrics are unique for an individual and they are very difficult to forget or steal as compared to other security mechanisms like password, pin, etc.
- Biometrics are more convenient, as users do not need remember these data.
- Biometrics are relatively more security as they are unique and cannot overlap for multiple users.
- Biometric systems based on behavioral data provide more secure authentication during an ongoing session.

Disadvantages of Biometrics

Biometrics also have certain disadvantages which are listed below –

- Biometrics are highly personalized data of an individual. Hence, they can have serious privacy concerns if mishandled.
- Biometric systems are relatively more complex and expensive.
- Biometric systems are sensitive to environmental conditions like dirty and moisturized area, dark area, etc. These factors can affect the operations of biometric systems.
- Some of the biometric techniques like fake fingerprints can be used for unauthorized access to systems.

Application of Biometrics

There are a number of applications where biometric systems are useful. Few of them are given below –

- Controlling workplace access
 - Biometric systems like fingerprint or iris scanners are used at offices, labs, or factories to allow only authorized employees to enter. It improves security and also helps in tracking attendance accurately.
- Identity establishment of people for authentic citizenship and immigration systems.
 - Biometric data (fingerprints, iris, face) is used in national ID systems (e.g., Aadhaar in India), passports, and visa processes. This ensures unique identification of each citizen or visitor and prevents identity fraud in immigration.
- Applying access control to sensitive information and systems.
 - Biometrics helps protect confidential digital systems, like financial databases, government portals, or hospital records. Only the authorized user (e.g., verified by fingerprint or face) can access the system, enhancing data security.
- Identifying criminals by forensics.
 - In forensic investigations, biometric evidence like fingerprints, facial images, or DNA can be matched against criminal databases to identify suspects, helping in solving crimes and tracking repeat offenders.
- Executing online e-commerce transactions.
 - Biometric authentication like fingerprint or face ID is now used to approve online payments or banking transactions. It adds an extra layer of security to protect users from fraud or unauthorized access.

- Fraud and theft reduction.
 - By verifying identity through biometric features, it becomes hard for someone to impersonate another person. This helps in reducing identity theft, financial fraud, and unauthorized access to secure systems.
- Law enforcement.
 - Law enforcement agencies use biometric databases (fingerprints, face recognition, iris scans) to track criminals, monitor suspects, and conduct background checks. It helps in maintaining public safety and order.

Automotive industry

There is an established relationship between eye movement and attention. Thus, tracking the car driver's eye movements can be very helpful in measuring the degree of sleepiness, tiredness, or drowsiness. The sleepiness of the driver can be detected by analyzing either blink duration and amplitude or the level of gaze activity.

Screen navigation:

One of the most important applications for people with disabilities is screen navigation. Using cameras, the application can track a person's eye movements in order to scroll a web page, write text, or perform actions by clicking on buttons on a computer or mobile devices. Therefore, this kind of application is gaining more attention recently due the rapid development and the growing need of new means of screen navigation especially on mobile devices platforms.

- Aviation:
 - The flight simulators track the pilot eye and head movement in order to analyze the pilot's behavior under realistic circumstances. This simulator is capable of evaluating a pilot's performance based on his eye movements combined with other information. It can be also used as an important training tool for new pilots in order to help them to look at the primary flight display (PFD) more regularly in order to monitor different airplane indicators.

Mobile

Mobile biometric solutions live at the intersection connectivity and identity. They integrate one or more biometric terms for authentication or identification purposes and take advantage of smartphones, tablets, other types of handhelds, wearable technology, and the Internet of things for versatile deployment capabilities. Thanks to the versatility brought by modern mobile technology, as well as the proliferation of

mobile paradigms in the consumer, public, and private world, mobile biometrics is becoming more and more important.

- Healthcare
 - In the field of healthcare, biometrics introduces an enhanced model. Medical records are among the most valuable personal documents; doctors need to be able to access them quickly, and they need to be accurate. A lack of security and good accounting can make the difference between timely and accurate diagnosis and health fraud.

Border control and airport

A key area of application for biometric technology is at the border. Biometric technology helps to automate the process of border crossing. Reliable and automated passenger screening initiatives and automated SAS (Smart Automated System) help to facilitate international passenger travel experience while improving the efficiency of government agencies and keeping borders safer than ever before.

AUTHENTICATION

Authentication is the process of verifying the identity of a user, device, or system to ensure that it is who or what it claims to be.

Strong authentication

Strong authentication is a security process that requires users to prove their identity using multiple, secure methods before they are granted access to a system, service, or physical space. It goes beyond just using a password by combining two or more independent factors that are hard to fake, steal, or guess.

Authentication Factors

People's identity are verified and identified by three basic means:

- Something they **have** (identity document or token)
- Something they **know** (password, PIN)
- Something they **are** (human body such as fingerprint or iris).

The strongest authentication involves a combination of all three.

Systems with more demanding requirements for security may use **location** and **time** as fourth and fifth factors. For example, users may be required to authenticate from specific locations, or during specific time windows.

Need for Strong authentication

Strong authentication is crucial in today's digital and security-driven world. It ensures that access to sensitive information, systems, or physical spaces is granted only to genuine and authorized users. Here are the key reasons why strong authentication is needed:

1. To Prevent Identity Theft and Fraud

Weak authentication methods (like passwords) are easily stolen, guessed, or hacked. Strong authentication — especially multi-factor authentication (MFA) or biometric verification — significantly reduces the risk of impersonation and unauthorized access.

2. To Secure Sensitive Information

Healthcare records, financial data, government databases, and personal accounts hold confidential data. Strong authentication helps protect this data from cyberattacks, leaks, or misuse.

3. To Enhance Trust in Online Transactions

Users and service providers need to trust that transactions (e.g., online banking, e-commerce, telemedicine) are conducted by legitimate users. Strong authentication builds trust and confidence in digital platforms.

4. To Ensure National Security

At borders, airports, defense facilities, and critical infrastructure, strong authentication (e.g., biometric checks) is vital to control access and prevent unauthorized intrusions or terrorism.

5. To Comply with Regulations

Many industries must comply with data protection laws (like GDPR, HIPAA, RBI guidelines, etc.) that mandate strong authentication to safeguard user privacy and data integrity.

6. To Prevent Insider Threats

Even within an organization, strong authentication ensures only the right employees access the right data, minimizing risks from internal misuse or accidental exposure.

7. To Support Secure Remote Access

With increasing remote work, strong authentication is critical to protect remote logins, VPN access, and cloud systems from unauthorized users.

Authentication Technologies

1. Password-based authentication

Password-based authentication is the most common form of authentication. Many apps and services require people to create passwords that use a combination of numbers, letters, and symbols to reduce the risk that a bad actor will guess them. However, passwords also create security and usability challenges. It's difficult for people to come up with and memorize a unique password for each of their online accounts, which is why they often reuse passwords. And attackers use many tactics to guess or steal passwords or lure people into sharing them unwillingly. For this reason, organizations are moving away from passwords to other more secure forms of authentication.

2. Certificate-based authentication

Certificate-based authentication is an encrypted method that enables devices and people to identify themselves to other devices and systems. Two common examples are a smart card or when an employee's device sends a digital certificate to a network or server.

3. Biometric authentication

In biometric authentication, people verify their identity using biological features. For example, many people use their finger or thumb to sign in to their phones, and some computers scan a person's face or retina to verify their identity. The biometric data is also linked to a specific device, so attackers can't use them without also gaining access to the device. This type of authentication is increasingly popular because it's easy for people—they don't have to memorize anything—and it's difficult for bad actors to steal, making it more secure than passwords.

5. One-time password

One-time passwords (OTP) are codes generated for a specific sign-in event that expire shortly after they're issued. They are delivered via SMS messages, email, or a hardware token.

6. Voice authentication

In voice authentication, the person trying to access a service receives a phone call, in which they're asked to enter a code or identify themselves verbally.

7. Multi-Factor Authentication

A higher level of confidence is provided by multi-factor authentication, which requires two or more login credentials. By supporting any quantity and configuration of verification factors, it expands upon two-factor authentication.

Several MFA techniques include:

Context-Based Authentication: This method looks for unusual logins by examining the user's location, IP address, and device identity in addition to other information.

Tracking user behavior such as mouse movements and keyboard patterns is known as behavioral biometrics.

Knowledge-based security question formulation.

Biometric authentication: facial recognition, iris scanning, or fingerprinting. MFA ensures that authorized users can access networks with ease while making it extremely difficult for hackers to simultaneously crack so many different forms of credentials. The extra stages, though, might not be as convenient for the user.

Protecting privacy and biometrics policy

Protecting Privacy in Biometrics

Biometric data (like fingerprints, facial scans, or iris patterns) is unique and permanent—if compromised, it can't be changed like a password. Protecting privacy means ensuring that users' biometric information is collected, stored, and used securely.

Key Privacy Protection Measures

Data Encryption

Data Encryption is the process of converting readable data (plaintext) into an unreadable format (ciphertext) so that only authorized parties can access it. It's one of the most important methods to protect sensitive information, including passwords, biometrics, financial data, and communications.

The methodology involves converting readable data, known as plaintext, into an unreadable format called ciphertext using mathematical algorithms and keys. This ensures that only authorized parties with the correct key can access the original data. The process begins with selecting an encryption algorithm, such as AES for symmetric encryption (same key for encryption and decryption) or RSA for asymmetric encryption (a public key for encryption and a private key for decryption).

The chosen key is then applied to the plaintext to generate ciphertext, which can be securely transmitted or stored. When needed, the ciphertext is decrypted back into plaintext using the appropriate key. This methodology protects sensitive information from unauthorized access, ensures confidentiality, and is widely used in securing communications, financial transactions, and biometric data.

Template Storage

In biometrics, template storage refers to how a user's biometric data (like fingerprints, facial features, or iris patterns) is securely stored after enrollment. Instead of saving the raw biometric image, the system processes it to create a biometric template—a mathematical representation containing unique features. This template is then used for future comparisons during authentication.

On-Device Processing – Whenever possible, process and store biometrics locally (e.g., on smartphones).

Minimization

Minimization in biometrics refers to the principle of collecting, storing, and processing only the minimum amount of biometric data necessary for a specific purpose. Since biometric data is highly sensitive and cannot be changed if compromised, limiting its use reduces privacy risks and the chances of misuse.

- Collect only required data
- Store only essential features
- Limit retention time
- Purpose-specific use
- On-device processing

Access Control

Biometric systems collect and store highly sensitive personal data – fingerprints, iris patterns, facial templates, voiceprints, etc. Because this data is unique and permanent, unauthorized access could lead to serious privacy violations. Access

control ensures that only authorized individuals, systems, or processes can view, modify, or use biometric data.

Anonymization & Tokenization

Biometric data (fingerprints, face templates, iris scans, etc.) is sensitive and permanent—unlike passwords, it can't be changed if compromised. To protect privacy and comply with data protection laws (like GDPR, HIPAA, and India's DPDP Act), biometric systems often avoid storing direct links between biometric templates and personal identity information (PII).

Anonymization is the process of removing or altering personally identifiable information (PII) so that an individual cannot be identified, either directly or indirectly, from the stored data.

Tokenization replaces sensitive biometric data or identifiers with a random, meaningless token (like a9xkL93p) that acts as a reference pointer.

Biometrics Policy

A biometrics policy defines the rules for collecting, storing, and using biometric data to ensure security, legal compliance, and user trust.

Core Elements of a Biometrics Policy

Purpose of Collection – Clearly state why biometric data is collected.

Consent – Obtain informed consent from users before collecting their data.

Data Retention – Define how long biometric data will be stored and when it will be deleted.

Compliance with Laws – Follow data protection regulations (like GDPR, HIPAA, or India's DPDP Act).

User Rights – Allow users to access, correct, or request deletion of their biometric data.

Breach Response Plan – Procedures in case biometric data is compromised.

PERFORMANCE MEASURES

Evaluation itself is an increasingly necessary step for biometrics market participation. Continuous performance evaluation allows the operator to improve and optimize the biometric system over time. It helps identify weaknesses and areas for enhancement while meeting the regulatory requirements for security and privacy.

The performance measurements of a biometric system are closely tied to False Reject Rate (FRR) and False Accept Rate (FAR).

FRR is also known as Type-I error or False Non Match Rate (FNMR) which states the likelihood of a legitimate user being rejected by the system.

FAR is referred to as Type-II error or False Match Rate (FMR) which states the likelihood of a false identity claim being accepted by the system.

An ideal biometric system is expected to produce zero value for both FAR and FRR. Means it should accept all genuine users and reject all fake identity claims, which is practically not achievable.

FAR and FRR are inversely proportional to each other. If FAR is improved, then the FRR declines. A biometric system providing high FRR ensures high security. If the FRR is too high, then the system requires to enter the live sample a number of times, which makes it less efficient.

The rate at which both the FAR and FRR are equal is the **Equal Error Rate (EER)**. It provides a value representing the point where both types of errors balance each other.

The performance of current biometrics technologies is far from the ideal. Hence the system developers need to keep a good balance between these two factors depending on the security requirements.

False Positive Identification Rate (FPIR) measures the probability that a biometric system incorrectly identifies an individual who is not in the database as being a match. The FPIR is approximately equal to the FMR in a one-to-one verification system multiplied by the number of entries in the database.

False Negative Identification Rate (FNIR) measures the probability that a system fails to correctly identify a registered user.

System performance indicator

Failure to Enroll (FER) - It is the ratio of the number of users that cannot be successfully enrolled in a biometric system and a total number of users presented to the biometric system.

Total number of attempts/Total number of enrollments

A lower value of FER indicates better population coverage.

Failure to capture (FTC) or Failure to acquire (FTA) – The rate at which the system is unable to capture a quality biometric sample based upon the number of attempts.

Acquired quality data/Total number of attempts

A lower value of FTA indicates better acquisition performance.

Ability to verify rate (AVR) – This describes the overall percentage of the particular end user population which can be enrolled in any kind of biometric system.

The AVR can be described as

$$AVR = [(1 - FER) * (1 - FRR)]$$

Where

FER- Failure to Enroll Rate.

FRR- False Rejection Rate.

